

SCXconvergence

SCX Industry Audit: Multi-Verifier Convergence Analysis Across Six Sectors

SCX

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Abstract

Abstract SCX/YajieManufacturingSemiconductorBankingCivil Servants Corethere exists——Civil Servants $M = 1$ BankingManufacturing $M > 1$ ISOJORC/NI 43-101 however $M > 1$ there exists $\sum g = 0$ i.e.allall $M > 1$ Conflict of Interestthere exists

Yajie“ $M > 1$ ”Banking/Manufacturing/ $M > 1$ $M > 1$ $\sum g = 0$ ——each“”“”
 $\sum g = 0$ SCX“”“”
 $\sum g = 0$ allSCX5,000/SCX1,000-2,500/

Abstract: This paper presents a systematic analysis of trust-mechanism restructuring under the SCX/Yajie Protocol across six critical industries: professional services, manufacturing, semiconductors, natural resources, banking, and civil service. Our central finding is a structural gradient in trust mechanisms: while services and civil service operate under a near- $M = 1$ self-declaration regime, banking, manufacturing, and resources already have formal $M > 1$ multi-verifier systems (Basel accords, ISO certification, JORC/NI 43-101 qualified-person regimes). However, all existing $M > 1$ systems share a fundamental defect: the absence of the $\sum g = 0$ mathematical constraint—the requirement that the sum of all verifier biases must equal zero and that all verifications must converge toward observable facts. Existing $M > 1$ systems employ multiple verifiers, but verifier biases may systematically lean in the same direction (regulatory capture, paid-auditor conflicts of interest); no mathematical mechanism forces convergence toward truth.

The revolutionary contribution of the Yajie Protocol is not “introducing $M > 1$ ” (banking/manufacturing/resources already have $M > 1$), but rather **superimposing the $\sum g = 0$ constraint on both existing and new $M > 1$ systems**—transforming each industry’s trust mechanism from “multiple verifiers whose biases may coexist” to “multiple verifiers whose biases must sum to zero.” $\sum g = 0$ constitutes SCX’s genuine technical contribution: it transforms verification from a

sociological question (“who is more credible?”) into a mathematical question (“do the biases converge?”).

Any industry participant that refuses multi-verifier audit under the $\sum g = 0$ constraint will have all its claims marked as **UNDECLARED**, rendering them non-tradable within the Yajie network. We conservatively estimate the global addressable SCX audit market across six industries at approximately \$500 billion/year, with SCX-capturable share projected at \$100–250 billion/year.

Keywords: SCX protocol, Yajie audit, multi-verifier convergence, $\sum g = 0$ constraint, industry audit, trust mechanism, verifier bias, audit sovereignty, information asymmetry, UNDECLARED marking.

1 / Introduction

1.1 / The Structural Defect of Trust

each however —Information Asymmetry

Definition 1.1. $\mathcal{A} = \{A_1, A_2, \dots, A_M\}$ $C \ M = 1 \ M > 1$

Definition 1.2. $g_i \in \mathbb{R} \ A_i \ g_i > 0 \ g_i < 0 \ g_i = 0$

Definition 1.3 ($\sum g = 0$). *Yajieall*

$$\sum_{i=1}^M g_i = 0 \tag{1}$$

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Table 1: / Current Verification Structures Across Six Industries

Industry	Claimant	Claim Content	$\frac{M}{\text{Current } M}$	$\frac{\sum g = 0}{\sum g = 0?}$
/ Services	/ Provider	/ Quality	1	/ No
Manufacturing	/ / Factory	/ Compliance	≈ 1	/ No
/ Manufac- turing				
Semiconductor	Wafer Fab /	/ Yield	1	/ No
/ Semicon- ductors	Fab			
/ Resources	/ Miner	/ Reserves	≈ 1	/ No
Banking	/ / Bank	/ CAR	> 1 ()	/ No
Banking				
Civil Servants	/ Govern-	/ Performance	1	/ No
/ Civil Ser- vice	ment			

1i.e.BankingManufacturing $M > 1$ ISOJORC $\sum g = 0$ —and—ISO $M > 1$ $\sum g = 0$

1.2 SCX/YajieCore / Core Innovation of SCX/Yajie

Yajie“ $M > 1$ ”——BankingManufacturingthere exists **all** $M > 1$ $\sum g = 0$ “”“”

Proposition 1.4 (convergence). $\sum g = 0$ $A_i g_i \neq 0$ $M - 1$ because

$$g_i = - \sum_{j \neq i} g_j \quad (2)$$

i.e.eachallon the contrary M

$\sum g = 0$ “”“” — SCX $\sum g = 0$ allUNDECLARED convergenceawaitingthere exists
each $\sum g = 0$

2 Formal Framework / Formal Framework

2.1 / Multi-Verifier Consensus Model

we formalizeSCX/YajieCore

Definition 2.1. $\mathcal{C} = \{C_1, C_2, \dots\}$ alleach C_k P_k $\theta_k \in \mathbb{R}$ for example θ_k

Definition 2.2. A_i C_k

$$\hat{\theta}_{k,i} = \theta_k + g_i + \varepsilon_{k,i} \quad (3)$$

where g_i A_i $\varepsilon_{k,i} \sim \mathcal{N}(0, \sigma_i^2)$

Definition 2.3 (Yajie). $\sum g = 0$ M C_k Yajie

$$\hat{\theta}_k^{Yajie} = \frac{1}{M} \sum_{i=1}^M \hat{\theta}_{k,i} \quad (4)$$

$\sum g = 0$ $\mathbb{E}[\hat{\theta}_k^{Yajie}] = \theta_k + \frac{1}{M} \sum_i g_i = \theta_k$ i.e. Yajie

Theorem 2.4 (Bias Detectability). $M \geq 3$ if A_i $|g_i| > 0$ there exists $1 - \alpha$

$$|g_i| > z_{\alpha/2} \cdot \frac{\sigma_i}{\sqrt{T}} \quad (5)$$

where T $z_{\alpha/2}$

theorem **2.4** $\sum g = 0$ such that “”—— T convergence

2.2 UNDECLARED / Game-Theoretic Analysis of UNDECLARED

Definition 2.5 (UNDECLARED). *if $P_k M > 1$ satisfying $\sum g = 0$ C_k SCX UNDECLAREDUNDECLARED SCX*

UNDECLARED “i.e.” Silence-as-Admission

Proposition 2.6 (i.e.). *SCXifthere existsYajie UNDECLARED Bayesian*

$$P(\mid UNDECLARED) > P() \tag{6}$$

there existsif and only ifUNDECLARED`”

UNDECLARED “” — UNDECLARED

2.3 / Audit Market Size Estimation

2SCX

Table 2: SCX / Potential SCX Audit Market by Industry

Industry	/ Global Rev- enue	/ Current Audit %	SCX Potential SCX Mar- ket
	~\$6.5	~0.5%	~\$325
Manufacturing	~\$17	~1.0%	~\$1,700
Semiconductor	~\$6,000	~2.0%	~\$120
	~\$5	~1.5%	~\$750
Banking	~\$6	~3.0%	~\$1,800
Civil Servants	~\$18	~0.3%	~\$540
	~\$53	—	~\$5,235

SCX5,000/ i.e.SCXwhere20%1,000/—

3 / Professional Services

3.1 Inflation / Current Structure: Inflation of Declarations

Core“”

- / **Consulting**BCGawaiting“” PPT There is no way to know“Client Renewal Rate”Quality ProofCorporate Inertia—Switching Costs

- / Legal Services “”
 — “” —there existsbecause
- / Advertising “” Growth —
- / Healthcare “”
- Higher Education / Higher Education “” Proof
 “” —i.e.

Akerlof—Information Asymmetrybecause

3.2 $\sum g = 0$ / What $\sum g = 0$ Exposes

SCX

- (1) / Decoupling of Brand Premium from Quality Premium $\sum g = 0$ “effectiveness” M Project12
 24
 SCX

- (2) / Case Selection Bias YajieCercisScore —80% “80%” 50%
- (3) / Measurability of Creative Output effectiveness M YajieScoreQuality Proof
- (4) / The Truth of Healthcare Quality “” ———YajieScore
- (5) “” / Exposure of Educational Value-Added SCX “” YajieScore—

3.3 / Winners and Losers

Table 3: SCX

/ Winners	/ Losers
Proof /Information Asymmetry	

3.4 / Recommended Strategy

- 0–12 SCXYajie “” i.e. $SCXM \geq 536$
 12–24 First MoversSCXdefineAudit DimensionsYajieScore “Yajie”
 24 YajieFirst Movers—→YajieScore→→

4 Manufacturing / Manufacturing

4.1 / Current Structure: False Security of Single-Point Inspection

Manufacturing assumption Internal Inspection awaiting Quality Assurance This assumption is wrong

Manufacturing there exists——ISO $\sum g = 0$ “”

(1) / Incentive Conflict

if Corruption——

(2) / Sampling Bias

(3) / Standard Relaxation define ——“”

for 100% for “”

4.2 $\sum g = 0$ / What $\sum g = 0$ Exposes

(1) reliability Scale / Magnitude of Self-Inspection Unreliability simultaneously- Internal Inspection $MM \geq 3g_i$ 0.1% 0.8%——“” Quality Integrity Deficit

(2) / Hidden Quality Risk Propagation in Supply Chains SCX each each YajieScore “” “”

(3) / Fundamental Flaw of Inspection Economics —— SCX “”

(4) ISO / The Ritualistic Nature of ISO Certification ISO 9001 awaiting $M > 1 \sum g = 0$ “” SCX ISO—— $\sum g = 0$

(5) / Separation of Process Quality from Inspection Quality

SCX “” “” ——

4.3 / Winners and Losers

Table 4: ManufacturingSCX

/ Winners	/ Losers
SPCTQM	“_”
	“”
	—

4.4 / Recommended Strategy

0–12SCX

+Internal Inspection

12–24Yajie allSCX “Yajie”YajieScore

24Yajie YajieCore “Yajie”

5 Semiconductor / Semiconductors

5.1 i.e. / Current Structure: Yield as State Secret

Semiconductor Core——yield——

Core TSMC3nmGAAGAAIntel 4

Wafer Fab“80%”——40%95%awaiting

——Core ifTSMCTSMC Semiconductor“”thus

“” “”

“”

5.2 $\sum g = 0$ ——Core / What $\sum g = 0$ Exposes——and the Core Tension

(1) $M\sum g = 0$ ——

(2) Semiconductor“” SCX“”

(3) Wafer FabeachAB SCXifWafer Fab

(4) “3.2%”—— $M > 1$ “”

Core vs. SemiconductorSCX SCXM > 1——SemiconductorCore Wafer Fabawait-
ingCompetitive Advantage

- **AZero-Knowledge AuditZK-Audit**——Zero-Knowledge ProofWafer FabSCXProof
ZKSemiconductor
- **BTEE**——Wafer FabTEE
- **C“”**——ifWafer FabSCXUNDECLARED
- **D**——Wafer FabSCXWafer Fab

5.3 / Three-Tier Audit Architecture

we propose

Table 5: Semiconductor / Three-Tier Semiconductor Audit Architecture

Level	Description	Application	Valuation
Level 1	Wafer Fab		
Level 2	Abstract		awaiting
Level 3	M > 1		

5.4 / Winners and Losers

Winner CategoriesTSMC——“Industry Consensus” “Provable Facts”Zero-Knowledge
ProofSemiconductorWafer Fab
Wafer FabInformation AsymmetryMarket Value

5.5 / Recommended Strategy

SemiconductorSCX“” “”
0–24 “Semiconductor” /ZKZK-AuditTEE-Auditfeasibility CoreSCX
24–48 Wafer Fab Level 1Level 2awaitingLevel 3
48 Level 3“” Level 1Wafer Fab“Level 3”

6 / Natural Resources

6.1 / Current Structure: What’s Underground, I Decide

——Coreall

- / Mineral Reserve Declarations “500LCE”
there existsCompany ValuationCEO——
- / The Classification Game “”(Proven)“(Probable)“(Possible) JORCNI 43-101
“” “”
- / Production Data —— “”
- / Fraudulence of Environmental Claims “” “” “” ——
- / Self-Certification of Carbon Credits “”
- / The Politics of Oil Reserves OPEC 1980OPEC“”——50%

6.2 $\sum g = 0$ / What $\sum g = 0$ Exposes

- (1) / Reserve Bubbles $M > 1$ “” $\sum g = 0$ if “Inflation”
- (2) Decays / The True Grade Decline Curve SCX $\sum g = 0$ “” ——
- (3) / Pre-Disaster Early Warning $M > 1$ if SCX—— “”
- (4) “” / The “Authenticity Deficit” of Carbon Credits Project SCX++ M if 60%40% “” Project
- (5) / The End of Reserve Politics if 1000SCXUNDECLAREDSCX “” “”

6.3 / Winners and Losers

Table 6: SCX

/ Winners	/ Losers
“Audit Premium”	Inflation
Project	
//	
	—

6.4 / Recommended Strategy

- 0–18 1–2SCX $M = 5$ i.e. ——
18–36 SCXalleach/SCXYajie
36Yajie “SCX” SCX-Certified CopperSCX-Certified Lithiumawaiting

7 Banking / Banking

7.1 / Current Structure: Self-Reporting Under Regulatory Rubber-Stamp

Banking assumption Both parts of this assumption have serious flaws

(1) / Banks' Self-Reporting Incentives

RWA — 1.2% 1.0% — “ ” — “ ”

(2) / Regulators' Verification Limitations i.e. ECB PRA Information Asymmetry

(3) / The Performativity of Stress Tests 2008 “ ” assumption GDP —

(4) auditability / The Unauditability of Complex Financial Instruments — CDO-CLO ETF — for there exists “ ”

(5) Banking / The Blind Spot of Systemic Risk — all simultaneously Fails A Bi.e.

7.2 $\sum g = 0$ / What $\sum g = 0$ Exposes

(1) $M > 1 \sum g = 0$ all

(2) “ ”

– “ ” Capital Integrity Deficit

(3) Non-Performing NPL “Non-Performing” define “ ” $SCX_M > 1$

(4) LCR NSFR “High-Quality Liquid Assets” define HQLA SCX reliability

(5) if SCX —

(6) if SCX Banking there exists “ ”

7.3 / Winners and Losers

Table 7: Banking SCX

/ Winners	/ Losers
SCX	
SCX	
stability	—

7.4 / Recommended Strategy

0–18 awaiting SCX RWA

18–36 Financial Regulation SCX SCX

36 YajieScore SCX “Yajie” Competitive Advantage——YajieScore

8 Civil Servants / Civil Service

8.1 $M = 1$ / Current Structure: The $M = 1$ Performance Loop

all Civil Servants Information Asymmetry——because

(1) $M = 1 \rightarrow \rightarrow \rightarrow$ “/” Corruption——

(2) **Corruption** Corruption/Corruption Corruption Corruption η ——because Corruption

(3) **effectiveness** “”

all “” “”——100%

(4)

8.2 $\sum g = 0$ / What $\sum g = 0$ Exposes

(1) Civil Servants $M > 1$ —— $\sum g = 0$ each if all $g_{\text{superior}} > 0$ ——i.e. “”

(2) **Score** AB BM > 1 if “YajieScore”

(3) **Corruption** Corruption SCX—— for example “” there exists $g - g > 0$ $\sum g = 0$ “Corruption YajieScore”

(4) “” SCX $M > 1$ NGO $\sum g = 0$ “Fails” will become provable — which is impossible under the current system

(5) **Scale** SCX——

(6) “XYProject”—— SCX $M > 1$ YajieScore

Core Civil Servants SCX all

- **ASCX**——Civil Servants SCX/Academic
- **B**——Corruption awaiting
- **C**——/simultaneously SCX

8.3 / Winners and Losers

Table 8: Civil ServantsSCX

/ Winners		/ Losers	
Civil Servants	Provable Loyalty	Competence	CorruptionRent-Seeking
			“”
			“”

8.4 / Recommended Strategy

0–24 SCX Corruption

24-48 SCXSCXCivil Servants/

48 SCX “SCX” SCX

9 / Cross-Industry Synthesis

9.1 $M = 1$ universality / The Universality of the $M = 1$ Problem

allCoreM = 1 although

Table 9: $M = 1$ / Unified View of the $M = 1$ Problem Across Six Industries

Industry	Claimant	Claim	Current Verifier
Manufacturing / Semiconductor	Wafer Fab	80% X	Wafer Fab
Banking / Civil Servants		15% /	

$M = 1$ — SCX/Yajiethere exists $M > 1$ therefore $M = 1$ “” —
all “” $M = 1$ — “” “”

9.2 $\sum g = 0$ / The Mathematical Violence of $\sum g = 0$

$\sum g = 0$ Yajie “”——

“Failsbecausebecause _____

$$\sum g = 0 \quad M \quad \sum g = 0 \text{---}$$

——because “” Quality Assurance

9.3 / Structural Sources of First-Mover Advantage

SCXFirst Movers——

- (1) / Data Accumulation Positive Feedback First Movers→→YajieScore→→→First Movers
- (2) / Standard-Setting Power First Movers Late MoversFirst MoversdefineAudit Dimensions
- (3) / Switching Cost Lock-in YajieScore“” Yajie——
- (4) / Talent and Auditor Lock-in / First Movers

9.4 UNDECLARED / The Deterrent Power of UNDECLARED

UNDECLARED SCX “” ——

SCXUNDECLARED awaiting“” UNDECLARED “Yajie” “”
UNDECLARED “i.e.” ——
SemiconductorCivil ServantsUNDECLARED //such thatUNDECLARED
textbf——“UNDECLARED” “”

9.5 / The New Economic Class of Auditors

SCXYajie

//—— $M = 1$ YajieCore

- $g_i \sum g = 0$
- “” ——Task
- Consistency
- SCX“” ——

9.6 / The Quantifiability of Trust Premium

SCX

SCX/Yajie

- YajieScore10%for
- SCXYajieScore= 0.95forScore= 0.70

- YajieAforB
- YajieScore20%for20%

SCX5–10 “”

9.7 / Cross-Industry Audit Linkages

SCXthere exists

- -SCX YajieScoreYajieScore
- -YajieScoreYajieScore
- -Civil Servants ifCivil ServantsSCXYajieScorethereforeSCX

SCX——

10 / Discussion: Risks and Challenges

10.1 / Auditor Collusion Risk

if $M\sum g = 0$ each g_i

- M ——
- ——
- “” ——

10.2 / Privacy-Transparency Balance

SemiconductorCivil Servants ZK-AuditTEE

10.3 / Audit Cost and Coverage

SCX —— for example“Scale” Audit Dimensions $\sum g = 0$ “”

10.4 / Geopolitical Resistance

SCX YajieScore

10.5 “False Positive” Corruption / False Positive Corruption Flagging Risk

Civil ServantsCorruption “Corruption” Civil Servants $p < 0.001$ ——Corruption

11 Conclusion / Conclusion and Action Framework

11.1 CoreConclusion / Core Conclusions

- (1) $\sum g = 0$ Civil Servants $M > 1$ BankingManufacturing $M > 1$ /ISO/JORC
SCX/Yajieall $\sum g = 0$
- (2) $\sum g = 0$ because“”—— Consistency
- (3) all —— First Movers
- (4) SemiconductorCivil Servants“”becauseSemiconductorCivil Servantsthere exists
ZK-AuditTEE-Audit
- (5) Yajie SCX5,000/SCX1,000–2,500/
- (6) UNDECLARED “i.e.”
- (7) SCXCore “Provable Integrity”

11.2 / Tiered Action Framework

Table 10: / Tiered Action Framework

Phase	Core Core Actions	Key Milestones
0–12	2–3 “” SCX	Yajie
12–36	Top 20	YajieScore
36+	Yajie “”	“” “” SCX

11.3 Concluding Outlook / Concluding Outlook

SCX/Yajie—— It does not change what industries do, but changes how industries**prove**
they did what they claimed
 SCX

“all $\sum g = 0$
 ——
 becausebecause”

A AppendixATerm / Appendix A: Glossary

Term / Term	define / Definition
M	$/M = 1M > 1$
g_i	$ig_i > 0g_i < 0$
$\sum g = 0$	YajieCoreall
YajieScore / Yajie Score	$\sum g = 0M$
Cercis	YajieScoreScore
UNDECLARED	SCXM > 1
/ Trust Premium	/
ZK-Audit	Zero-Knowledge AuditProofsatisfying
TEE-Audit	
/ Capital Integrity Deficit	
/ Quality Integrity Deficit	
Inflation / Reserve Inflation Factor	
i.e. / Silence-as- Admission	there existsUNDECLARED

Table 11: Term / Glossary of Terms

B AppendixBAudit Dimensions / Appendix B: Industry Audit Dimensions Reference

B.1 Audit Dimensions / Services Audit Dimensions

CoreAudit Dimensions		M	
Higher Edu- cation	-	5+	Project12–24
		5+	
		5+	Project
		7+	
		5+	

B.2 ManufacturingAudit Dimensions / Manufacturing Audit Dimensions

CoreAudit Dimensions		M	
stability	vs	3+	
	Cpk	3+	
		3+	
	YajieScore	N/A	

B.3 SemiconductorAudit Dimensions / Semiconductor Audit Dimensions

CoreAudit Dimensions		Di-	
<5nm ≥7nm	stability	Level 12	TEE-Audit
		Level 23	+
		Level 3	

B.4 Audit Dimensions / Resources Audit Dimensions

CoreAudit Dimensions		M	
	/	5+	+
		5+	+
		5+	/
		5+	

B.5 BankingAudit Dimensions / Banking Audit Dimensions

CoreAudit Dimensions	M
RWANPL	5+
VaR	5+
LCRNSFR	5+
	3+

B.6 Civil ServantsAudit Dimensions / Civil Service Audit Dimensions

CoreAudit Dimensions	M	
	5+	Project
	5+	
	5+	
	5+	/
	7+	
Corruption -	7+	

C AppendixC / Appendix C: Key Mathematical Derivations

C.1 Bias DetectabilityProof / Proof of Bias Detectability (Theorem 2.4)

Proof. $A_i T$

$$\hat{\theta}_{k,i} = \theta_k + g_i + \varepsilon_{k,i}, \quad \varepsilon_{k,i} \sim \mathcal{N}(0, \sigma_i^2) \quad (7)$$

$$\sum g = 0M - 1$$

$$\bar{\theta}_{k,-i} = \frac{1}{M-1} \sum_{j \neq i} \hat{\theta}_{k,j} = \theta_k + \frac{1}{M-1} \sum_{j \neq i} g_j + \bar{\varepsilon}_{k,-i} \quad (8)$$

where $\bar{\varepsilon}_{k,-i} \sim \mathcal{N}(0, \sigma_{-i}^2 / (M-1))$ $\sigma_{-i}^2 = \frac{1}{M-1} \sum_{j \neq i} \sigma_j^2$

define

$$D_{k,i} = \hat{\theta}_{k,i} - \bar{\theta}_{k,-i} = g_i - \frac{1}{M-1} \sum_{j \neq i} g_j + (\varepsilon_{k,i} - \bar{\varepsilon}_{k,-i}) \quad (9)$$

$\sum g = 0$ $g_i = - \sum_{j \neq i} g_j$ therefore

$$\frac{1}{M-1} \sum_{j \neq i} g_j = -\frac{g_i}{M-1} \quad (10)$$

$$D_{k,i} = g_i + \frac{g_i}{M-1} + \eta_{k,i} = \frac{M}{M-1}g_i + \eta_{k,i} \quad (11)$$

where $\eta_{k,i} \sim \mathcal{N}(0, \sigma_i^2 + \sigma_{-i}^2/(M-1))$

$$\bar{D}_i = \frac{1}{T} \sum_{k=1}^T D_{k,i} = \frac{M}{M-1}g_i + \bar{\eta}_i, \quad \bar{\eta}_i \sim \mathcal{N}\left(0, \frac{\sigma_i^2 + \sigma_{-i}^2/(M-1)}{T}\right) \quad (12)$$

$H_0 : g_i = 0$

$$Z = \frac{\bar{D}_i}{\sqrt{(\sigma_i^2 + \sigma_{-i}^2/(M-1))/T}} \sim \mathcal{N}(0, 1) \quad (13)$$

$H_0 : |Z| > z_{\alpha/2}$ i.e.

$$\left| \frac{M}{M-1}g_i \right| > z_{\alpha/2} \cdot \sqrt{\frac{\sigma_i^2 + \sigma_{-i}^2/(M-1)}{T}} \quad (14)$$

$T \rightarrow \infty \rightarrow 0$ therefore any g_i

□

C.2 i.e. / Formal Game Model of Silence-as-Admission

Proof. $P(\tau \in \{H, L\}) = 1$ where $H \iff L$ $P(\tau = H) = p, P(\tau = L) = 1 - p$

$P(\text{SCX} | \tau = H) = 1$ if $\tau = L$

$P(\text{HR}_H | \tau = L) = 0$

$H \text{ a } L \text{ r } r$

$$P(\tau = L | r) = \frac{P(r | L) \cdot P(L)}{P(r | L) \cdot P(L) + P(r | H) \cdot P(H)} = \frac{1 \cdot (1 - p)}{1 \cdot (1 - p) + 0 \cdot p} = 1 \quad (15)$$

therefore $P(\text{UNDECLARED} | r) = 1 > P(\text{UNDECLARED}) = 1 - p$ for any $p < 1$

□

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